

Info:
 profile: CS. Columbia.edu/~gtchuang/discrete/index.html
 email: gtchuang@columbia.edu

Today: - mathematical stats
 - logic & quantification
 - course & logistics

- 1.) make precise stats
- 2.) Argue that proofs are true/false

Propositions: T/F statements

Def Quantify: when we want to use a variable, specify it is from.

Common Sets: Integers $\{0, 1, 2, 3, -1, -2, -3, \dots\}$

Reals: $\mathbb{R}$
 $x \in \mathbb{R}$

Natural #'s $\{1, 2, 3, \dots\} \mathbb{N}$

Rational #'s $\{\frac{1}{2}, \frac{2}{3}, 0, \dots\}$

Universal Quantifier: $\forall$

ex: for all $x \in \mathbb{Z}$
 $x^2 = x$
 for all $x \in \mathbb{Z}$
 x is div by 7

Existence quantifier

There Exists "existential" $\exists$
 $\exists x \in \mathbb{Z}$
 $x^2 \geq x$

Truth Table And, Or Not operators

P	Q	P AND Q
T	T	T
T	F	F
F	T	F
F	F	F

implies, "if P then Q"

$P \Rightarrow Q$ has the statement broken its promise.

P	Q	$P \Rightarrow Q$
T	T	T
F	F	T
T	F	F
F	T	T

$P \Rightarrow Q$ is true only if it's not possible for $P = T, Q = F$

if and only if $P \Leftrightarrow Q$

$P \Leftrightarrow Q$ AND $Q \Leftrightarrow P$

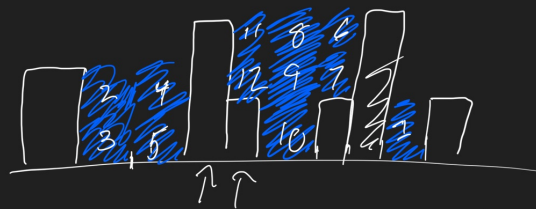
P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	F	T
F	T	F

Ex $x=5 \Leftrightarrow x+2=7$
 T

Sum: waiter

Sum = 1 = 3 = 5 = 8 = 9 = 10

1 < right



1, r



if

else

area = 7

(5, 4)
(4, 2)
(2, 2)
(0, 1)

$2 \cdot 4 = 8$

$5 > 1 \wedge 5 > 2$

$3 > 1 \wedge 3 < 2$

$3 > 1$

